PQC DAILY STUDY SERIES

Performance Overhead of PQC in Real-World Protocols

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Understanding PQC Performance Overhead

As the world transitions to Post-Quantum Cryptography (PQC), one of the most critical considerations is performance. While offering security against quantum threats, PQC algorithms inherently come with different performance characteristics compared to their classical counterparts (like RSA and ECC). This overhead can manifest as increased latency, higher computational load, and larger key/signature sizes, impacting the user experience and system efficiency in internet protocols.

Why Performance Matters

In protocols like TLS, SSH, and IKEv2 (used in IPsec VPNs), every millisecond counts. Handshake delays can lead to slower page loads, sluggish remote connections, and reduced throughput. For large-scale services, even a minor increase in CPU load per connection can translate to significant infrastructure costs. Therefore, understanding and mitigating this overhead is key to successful PQC adoption.

Key Areas of Performance Impact

- **Latency:** PQC handshakes, especially those involving larger keys and more complex computations, can take longer than classical handshakes. Hybrid modes (running a classical and a PQC algorithm together) further add to this, as they require two sets of cryptographic operations.
- **Computational Cost:** Algorithms like ML-KEM and ML-DSA are generally efficient, but they can still be more CPU-intensive than highly optimized classical algorithms like X25519. This is a particular concern for servers handling thousands of simultaneous connections or for resource-constrained IoT devices.
- **Bandwidth Usage:** PQC public keys and signatures are often significantly larger than those in ECC. For example, an ML-KEM-768 public key is over 1KB, whereas an ECC P-256 key is only 65 bytes. This increases the amount of data exchanged during a handshake, which can be a problem for networks with limited bandwidth.

Performance Comparison: Classical vs. PQC

The table below provides a simplified comparison of key metrics for algorithms commonly used in TLS handshakes. Actual performance can vary significantly based on the implementation, hardware, and protocol.

Algorithm	Type	Public Key Size (bytes)	Ciphertext/Signature Size (bytes)	Relative Computational Cost
ECDH (P-256)	Classical KEM	~65	~65	Low
ML-KEM-768	PQC KEM	~1184	~1088	Medium
ECDSA (P-256)	Classical Signature	~65	~72	Low
ML-DSA-65	PQC Signature	~1952	~3296	Medium-High

> **Real-World Applicability: Mitigating the Overhead**
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> Engineers are actively working on strategies to minimize PQC's performance impact. These include optimizing cryptographic libraries for specific hardware (using AVX2/NEON instructions), using hybrid modes only where necessary, and selecting PQC algorithms that offer the best balance of security and performance for a given use case. For instance, stateless protocols like QUIC (HTTP/3) can amortize the cost of the initial PQC handshake over a long-lived connection, making the overhead less noticeable over time.

Daily Quiz

1. What are the three main areas of performance impact when deploying PQC?
2. Why is the size of PQC public keys and signatures a concern for some applications?
3. What is a "hybrid mode" in the context of PQC deployment, and how does it affect performance?